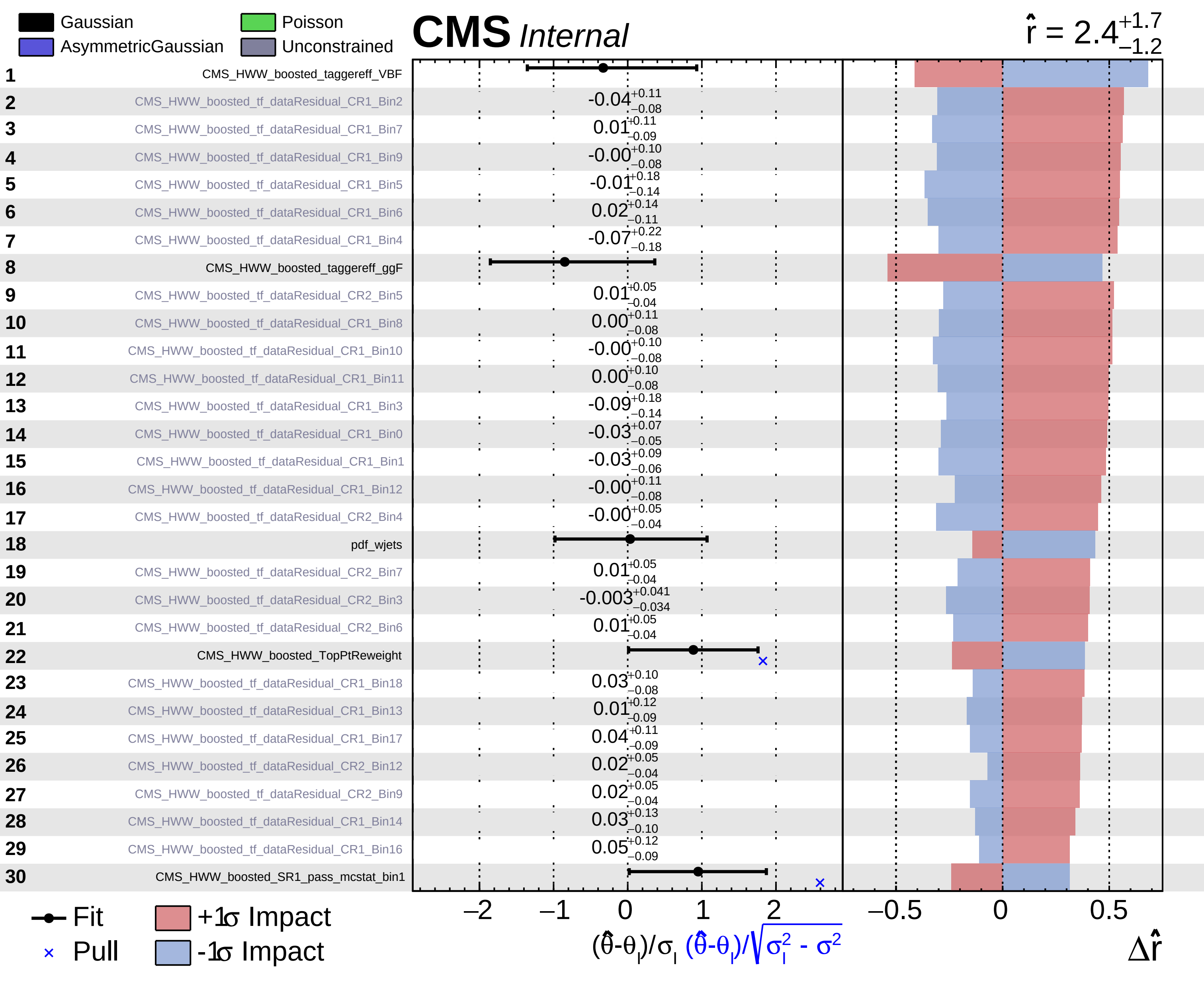
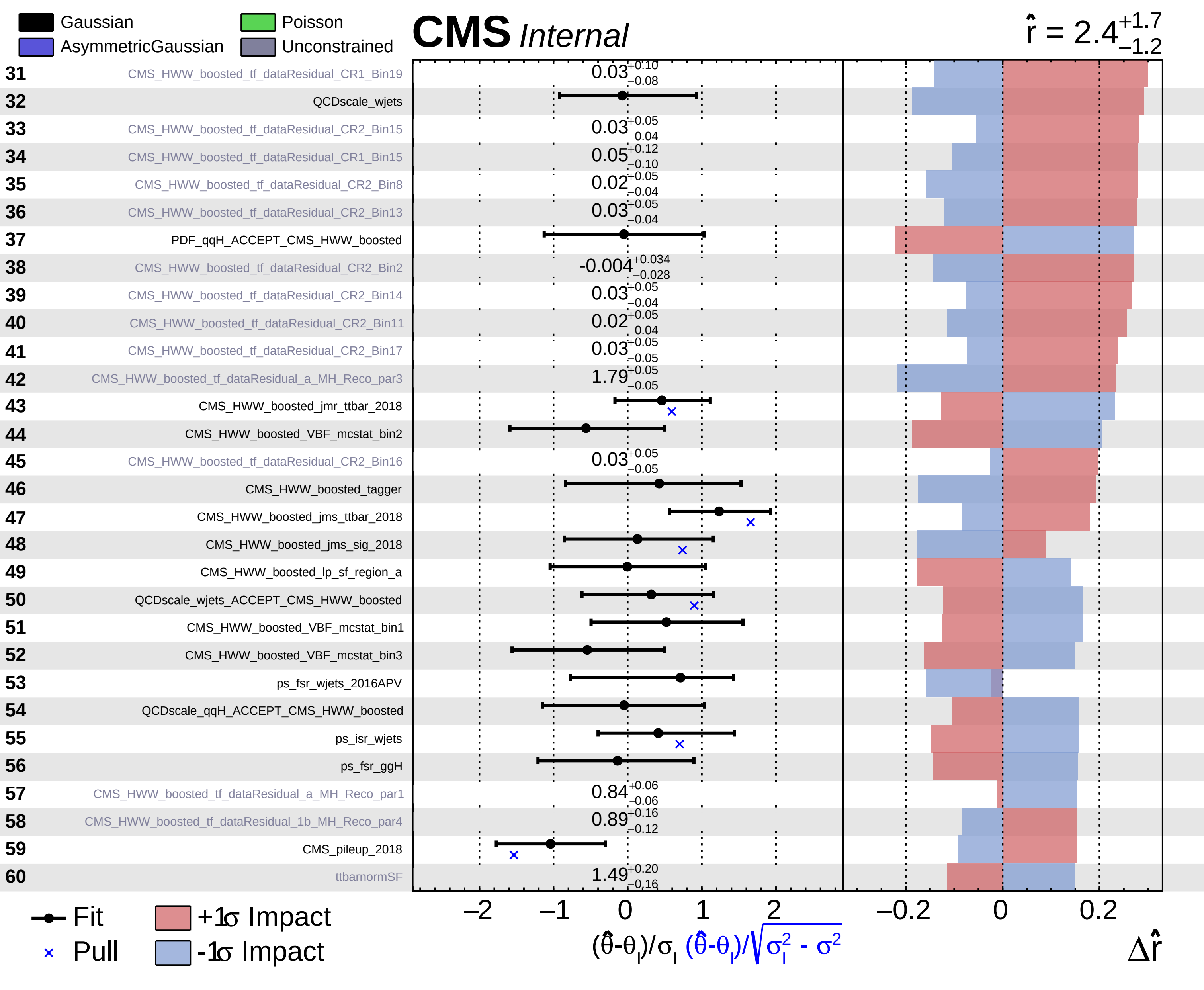
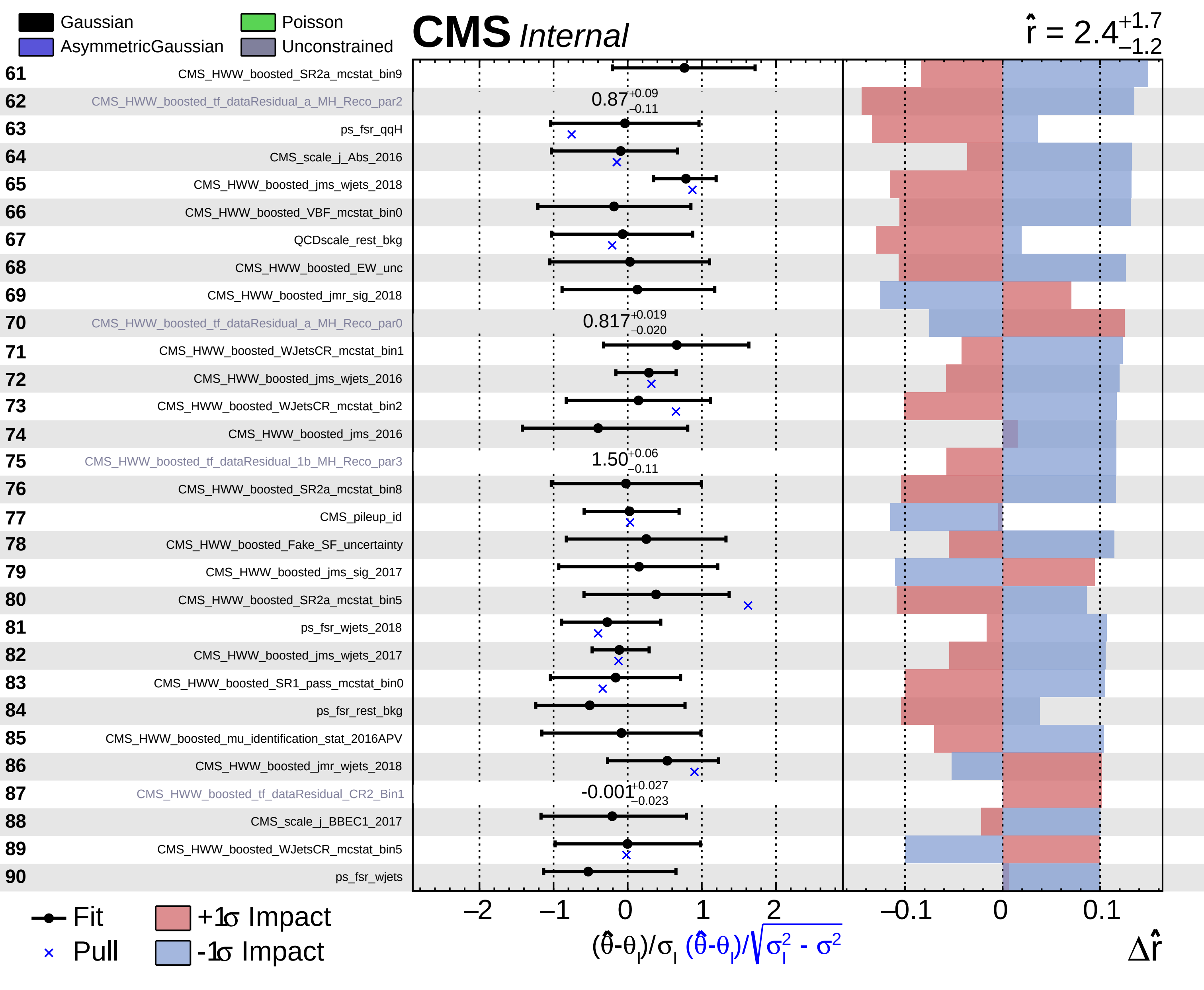
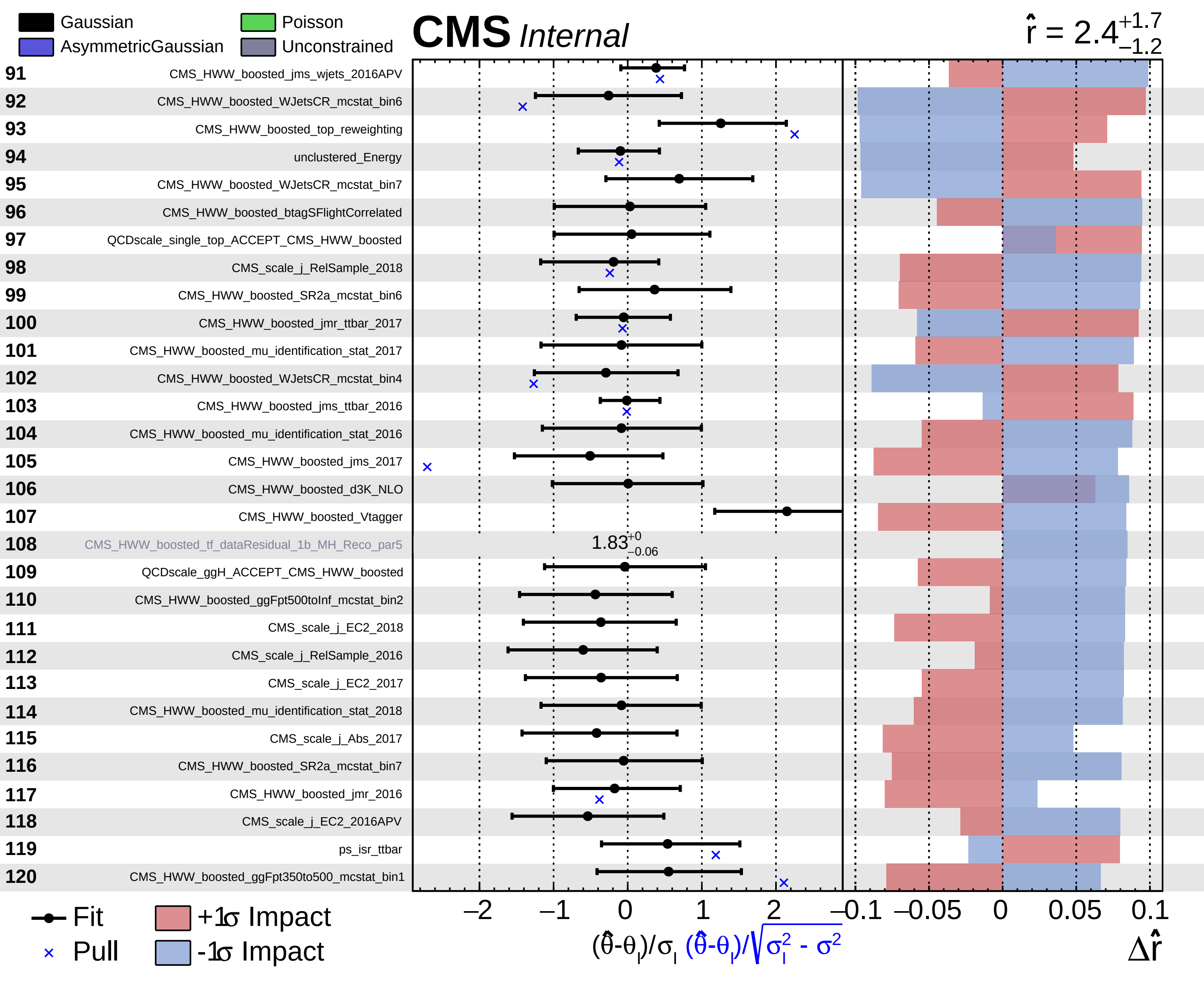


Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Internal
r̂ = 2.4 +1.7 −1.2
1 CMS_HWW_boosted_taggereff_VBF
2 CMS_HWW_boosted_tf_dataResidual_CR1_Bin2 -0.04 +0.11 −0.08
3 CMS_HWW_boosted_tf_dataResidual_CR1_Bin7 0.01 +0.11 −0.09
4 CMS_HWW_boosted_tf_dataResidual_CR1_Bin9 -0.00 +0.10 −0.08
5 CMS_HWW_boosted_tf_dataResidual_CR1_Bin5 -0.01 +0.18 −0.14
6 CMS_HWW_boosted_tf_dataResidual_CR1_Bin6 0.02 +0.14 −0.11
7 CMS_HWW_boosted_tf_dataResidual_CR1_Bin4 -0.07 +0.22 −0.18
8 CMS_HWW_boosted_taggereff_ggF
9 CMS_HWW_boosted_tf_dataResidual_CR2_Bin5 0.01 +0.05 −0.04
10 CMS_HWW_boosted_tf_dataResidual_CR1_Bin8 0.00 +0.11 −0.08
11 CMS_HWW_boosted_tf_dataResidual_CR1_Bin10 -0.00 +0.10 −0.08
12 CMS_HWW_boosted_tf_dataResidual_CR1_Bin11 0.00 +0.10 −0.08
13 CMS_HWW_boosted_tf_dataResidual_CR1_Bin3 -0.09 +0.18 −0.14
14 CMS_HWW_boosted_tf_dataResidual_CR1_Bin0 -0.03 +0.07 −0.05
15 CMS_HWW_boosted_tf_dataResidual_CR1_Bin1 -0.03 +0.09 −0.06
16 CMS_HWW_boosted_tf_dataResidual_CR1_Bin12 -0.00 +0.11 −0.08
17 CMS_HWW_boosted_tf_dataResidual_CR2_Bin4 -0.00 +0.05 −0.04
18 pdf_wjets
19 CMS_HWW_boosted_tf_dataResidual_CR2_Bin7 0.01 +0.05 −0.04
20 CMS_HWW_boosted_tf_dataResidual_CR2_Bin3 -0.003 +0.041 −0.034
21 CMS_HWW_boosted_tf_dataResidual_CR2_Bin6 0.01 +0.05 −0.04
22 CMS_HWW_boosted_TopPtReweight
23 CMS_HWW_boosted_tf_dataResidual_CR1_Bin18 0.03 +0.10 −0.08
24 CMS_HWW_boosted_tf_dataResidual_CR1_Bin13 0.01 +0.12 −0.09
25 CMS_HWW_boosted_tf_dataResidual_CR1_Bin17 0.04 +0.11 −0.09
26 CMS_HWW_boosted_tf_dataResidual_CR2_Bin12 0.02 +0.05 −0.04
27 CMS_HWW_boosted_tf_dataResidual_CR2_Bin9 0.02 +0.05 −0.04
28 CMS_HWW_boosted_tf_dataResidual_CR1_Bin14 0.03 +0.13 −0.10
29 CMS_HWW_boosted_tf_dataResidual_CR1_Bin16 0.05 +0.12 −0.09
30 CMS_HWW_boosted_SR1_pass_mcstat_bin1
−2 −1 0 1 2
−0.5 0 0.5
Fit
Pull
+1σ Impact
-1σ Impact
(θ̂-θ_I)/σ_I (θ̂-θ_I)/√(σ_I² - σ²)
Δr̂

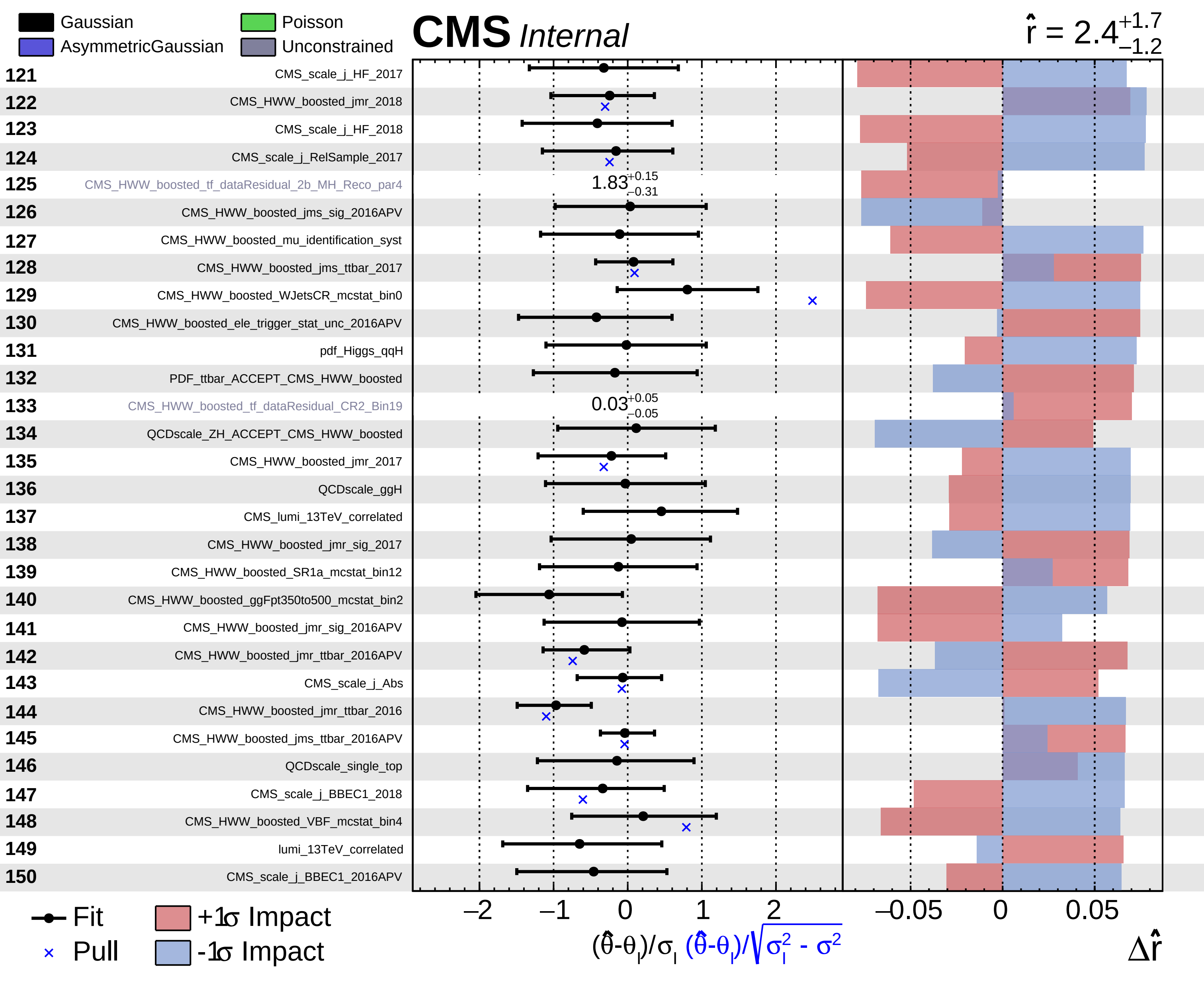

Gaussian
AsymmetricGaussian
Poisson
Unconstrained
CMS Internal
$\hat{r} = 2.4^{+1.7}_{-1.2}$
31 CMS_HWW_boosted_tf_dataResidual_CR1_Bin19 $0.03^{+0.10}_{-0.08}$
32 QCDscale_wjets
33 CMS_HWW_boosted_tf_dataResidual_CR2_Bin15 $0.03^{+0.05}_{-0.04}$
34 CMS_HWW_boosted_tf_dataResidual_CR1_Bin15 $0.05^{+0.12}_{-0.10}$
35 CMS_HWW_boosted_tf_dataResidual_CR2_Bin8 $0.02^{+0.05}_{-0.04}$
36 CMS_HWW_boosted_tf_dataResidual_CR2_Bin13 $0.03^{+0.05}_{-0.04}$
37 PDF_qqH_ACCEPT_CMS_HWW_boosted
38 CMS_HWW_boosted_tf_dataResidual_CR2_Bin2 $-0.004^{+0.034}_{-0.028}$
39 CMS_HWW_boosted_tf_dataResidual_CR2_Bin14 $0.03^{+0.05}_{-0.04}$
40 CMS_HWW_boosted_tf_dataResidual_CR2_Bin11 $0.02^{+0.05}_{-0.04}$
41 CMS_HWW_boosted_tf_dataResidual_CR2_Bin17 $0.03^{+0.05}_{-0.05}$
42 CMS_HWW_boosted_tf_dataResidual_a_MH_Reco_par3 $1.79^{+0.05}_{-0.05}$
43 CMS_HWW_boosted_jmr_ttbar_2018
44 CMS_HWW_boosted_VBF_mcstat_bin2
45 CMS_HWW_boosted_tf_dataResidual_CR2_Bin16 $0.03^{+0.05}_{-0.05}$
46 CMS_HWW_boosted_tagger
47 CMS_HWW_boosted_jms_ttbar_2018
48 CMS_HWW_boosted_jms_sig_2018
49 CMS_HWW_boosted_lp_sf_region_a
50 QCDscale_wjets_ACCEPT_CMS_HWW_boosted
51 CMS_HWW_boosted_VBF_mcstat_bin1
52 CMS_HWW_boosted_VBF_mcstat_bin3
53 ps_fsr_wjets_2016APV
54 QCDscale_qqH_ACCEPT_CMS_HWW_boosted
55 ps_isr_wjets
56 ps_fsr_ggH
57 CMS_HWW_boosted_tf_dataResidual_a_MH_Reco_par1 $0.84^{+0.06}_{-0.06}$
58 CMS_HWW_boosted_tf_dataResidual_1b_MH_Reco_par4 $0.89^{+0.16}_{-0.12}$
59 CMS_pileup_2018
60 ttbarnormSF $1.49^{+0.20}_{-0.16}$
−2
−1
0
1
2
−0.2
0
0.2
Fit
Pull
+1σ Impact
−1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$
$(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$



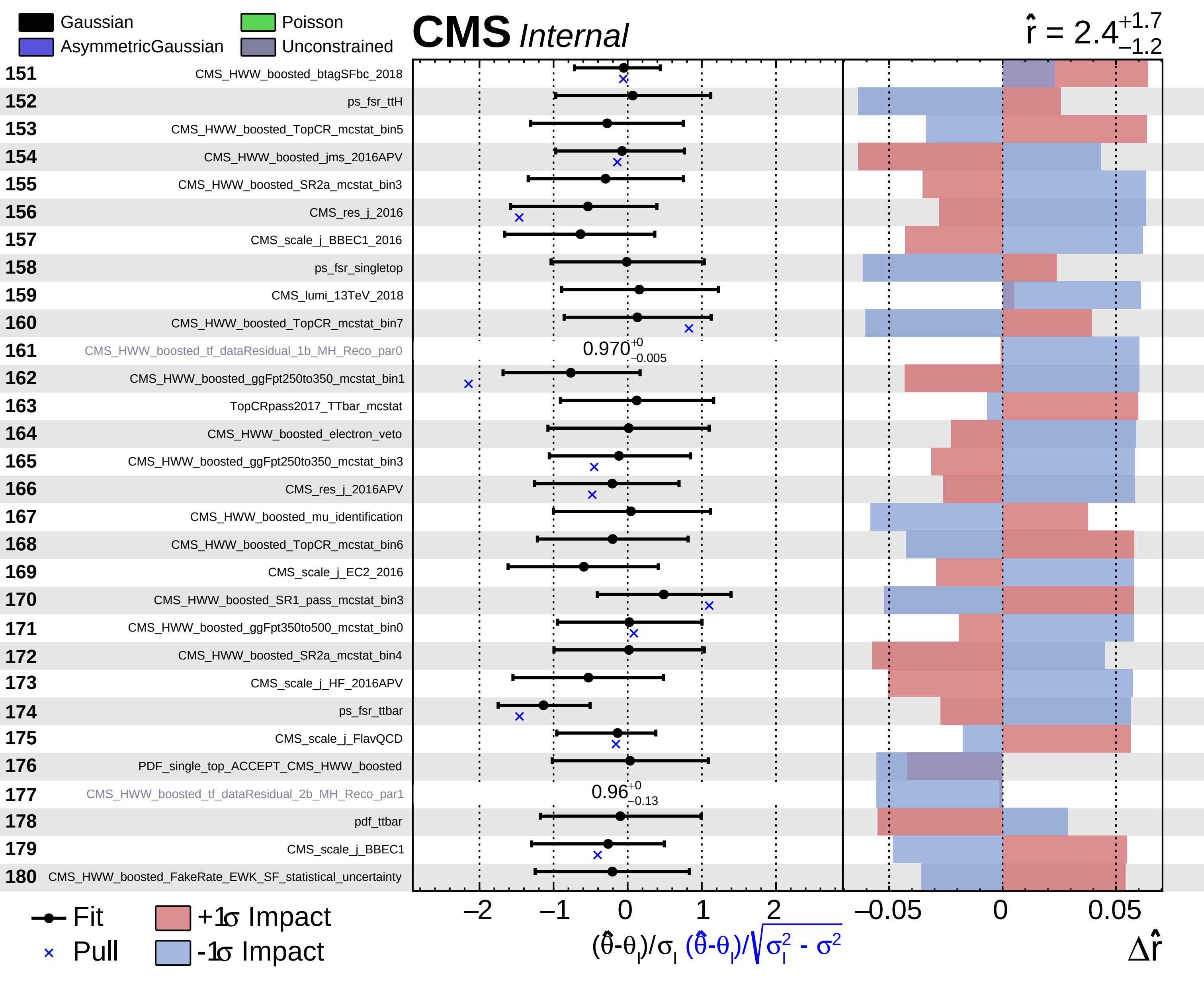
Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Internal
$\hat{r} = 2.4^{+1.7}_{-1.2}$
61 CMS_HWW_boosted_SR2a_mcstat_bin9
62 CMS_HWW_boosted_tf_dataResidual_a_MH_Reco_par2
$0.87^{+0.09}_{-0.11}$
63 ps_fsr_qqH
64 CMS_scale_j_Abs_2016
65 CMS_HWW_boosted_jms_wjets_2018
66 CMS_HWW_boosted_VBF_mcstat_bin0
67 QCDscale_rest_bkg
68 CMS_HWW_boosted_EW_unc
69 CMS_HWW_boosted_jmr_sig_2018
70 CMS_HWW_boosted_tf_dataResidual_a_MH_Reco_par0
$0.817^{+0.019}_{-0.020}$
71 CMS_HWW_boosted_WJetsCR_mcstat_bin1
72 CMS_HWW_boosted_jms_wjets_2016
73 CMS_HWW_boosted_WJetsCR_mcstat_bin2
74 CMS_HWW_boosted_jms_2016
75 CMS_HWW_boosted_tf_dataResidual_1b_MH_Reco_par3
$1.50^{+0.06}_{-0.11}$
76 CMS_HWW_boosted_SR2a_mcstat_bin8
77 CMS_pileup_id
78 CMS_HWW_boosted_Fake_SF_uncertainty
79 CMS_HWW_boosted_jms_sig_2017
80 CMS_HWW_boosted_SR2a_mcstat_bin5
81 ps_fsr_wjets_2018
82 CMS_HWW_boosted_jms_wjets_2017
83 CMS_HWW_boosted_SR1_pass_mcstat_bin0
84 ps_fsr_rest_bkg
85 CMS_HWW_boosted_mu_identification_stat_2016APV
86 CMS_HWW_boosted_jmr_wjets_2018
87 CMS_HWW_boosted_tf_dataResidual_CR2_Bin1
$-0.001^{+0.027}_{-0.023}$
88 CMS_scale_j_BBEC1_2017
89 CMS_HWW_boosted_WJetsCR_mcstat_bin5
90 ps_fsr_wjets
−2
−1
0
1
2
−0.1
0
0.1
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$
$(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$



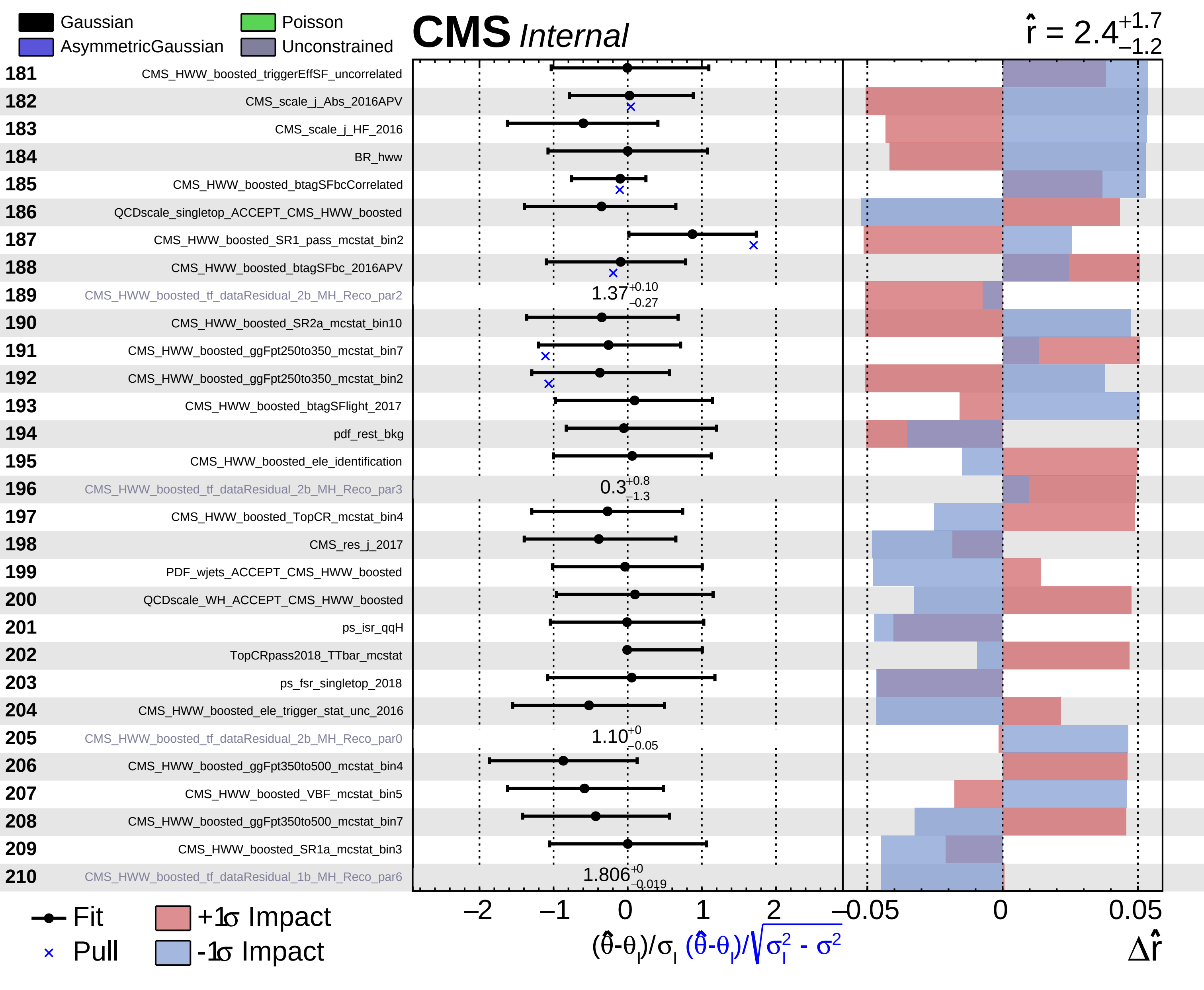
Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Internal
r̂ = 2.4 +1.7 −1.2
91 CMS_HWW_boosted_jms_wjets_2016APV
92 CMS_HWW_boosted_WJetsCR_mcstat_bin6
93 CMS_HWW_boosted_top_reweighting
94 unclustered_Energy
95 CMS_HWW_boosted_WJetsCR_mcstat_bin7
96 CMS_HWW_boosted_btagSFlightCorrelated
97 QCDscale_single_top_ACCEPT_CMS_HWW_boosted
98 CMS_scale_j_RelSample_2018
99 CMS_HWW_boosted_SR2a_mcstat_bin6
100 CMS_HWW_boosted_jmr_ttbar_2017
101 CMS_HWW_boosted_mu_identification_stat_2017
102 CMS_HWW_boosted_WJetsCR_mcstat_bin4
103 CMS_HWW_boosted_jms_ttbar_2016
104 CMS_HWW_boosted_mu_identification_stat_2016
105 CMS_HWW_boosted_jms_2017
106 CMS_HWW_boosted_d3K_NLO
107 CMS_HWW_boosted_Vtagger
108 CMS_HWW_boosted_tf_dataResidual_1b_MH_Reco_par5
1.83 +0 −0.06
109 QCDscale_ggH_ACCEPT_CMS_HWW_boosted
110 CMS_HWW_boosted_ggFpt500toInf_mcstat_bin2
111 CMS_scale_j_EC2_2018
112 CMS_scale_j_RelSample_2016
113 CMS_scale_j_EC2_2017
114 CMS_HWW_boosted_mu_identification_stat_2018
115 CMS_scale_j_Abs_2017
116 CMS_HWW_boosted_SR2a_mcstat_bin7
117 CMS_HWW_boosted_jmr_2016
118 CMS_scale_j_EC2_2016APV
119 ps_isr_ttbar
120 CMS_HWW_boosted_ggFpt350to500_mcstat_bin1
Fit
Pull
+1σ Impact
-1σ Impact
−2
−1
0
1
2
−0.1
−0.05
0
0.05
0.1
(θ̂-θ_I)/σ_I (θ̂-θ_I)/√(σ_I² - σ²)
Δr̂

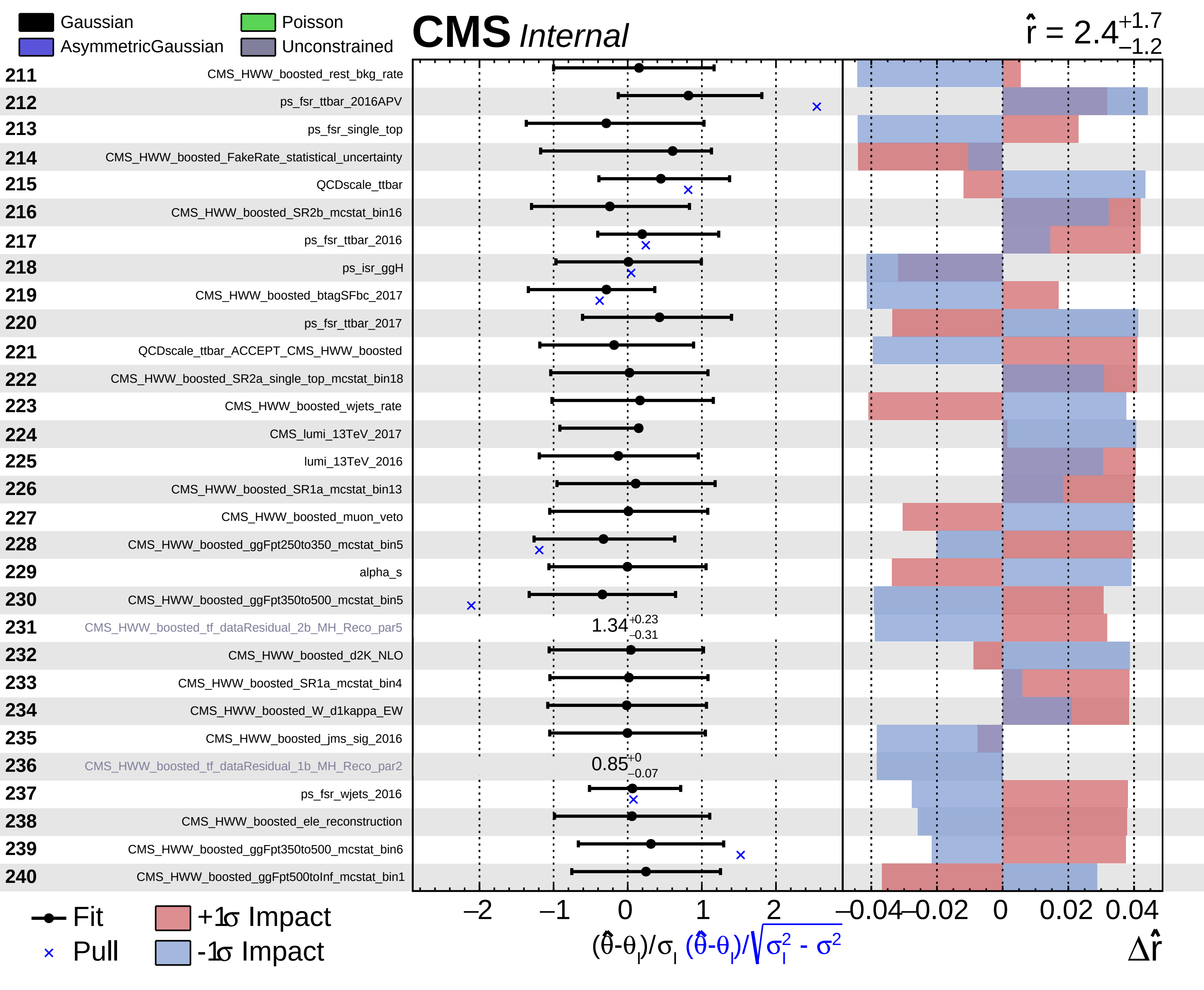


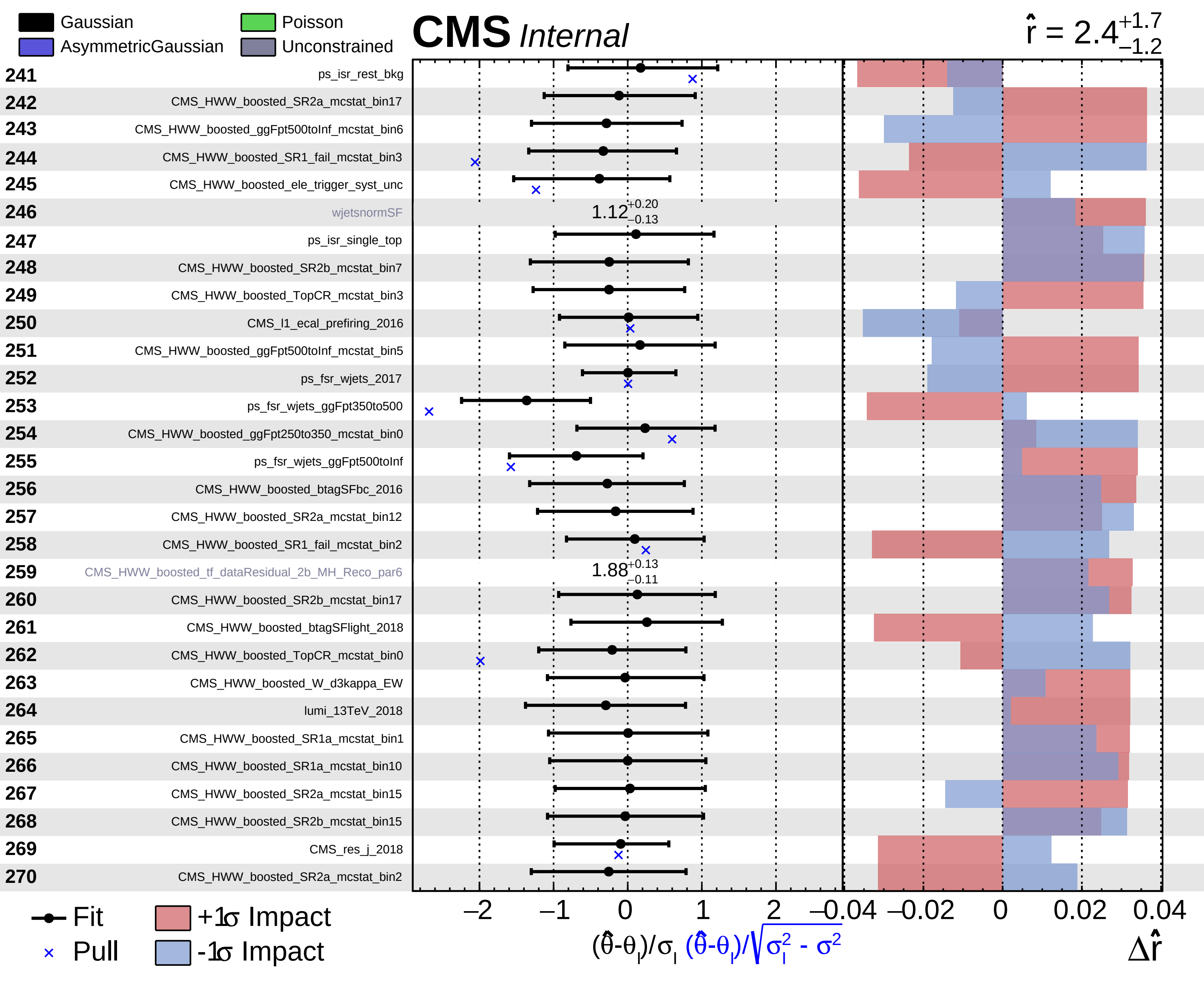
Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Internal
$\hat{r} = 2.4^{+1.7}_{-1.2}$
121 CMS_scale_j_HF_2017
122 CMS_HWW_boosted_jmr_2018
123 CMS_scale_j_HF_2018
124 CMS_scale_j_RelSample_2017
125 CMS_HWW_boosted_tf_dataResidual_2b_MH_Reco_par4
$1.83^{+0.15}_{-0.31}$
126 CMS_HWW_boosted_jms_sig_2016APV
127 CMS_HWW_boosted_mu_identification_syst
128 CMS_HWW_boosted_jms_ttbar_2017
129 CMS_HWW_boosted_WJetsCR_mcstat_bin0
130 CMS_HWW_boosted_ele_trigger_stat_unc_2016APV
131 pdf_Higgs_qqH
132 PDF_ttbar_ACCEPT_CMS_HWW_boosted
133 CMS_HWW_boosted_tf_dataResidual_CR2_Bin19
$0.03^{+0.05}_{-0.05}$
134 QCDscale_ZH_ACCEPT_CMS_HWW_boosted
135 CMS_HWW_boosted_jmr_2017
136 QCDscale_ggH
137 CMS_lumi_13TeV_correlated
138 CMS_HWW_boosted_jmr_sig_2017
139 CMS_HWW_boosted_SR1a_mcstat_bin12
140 CMS_HWW_boosted_ggFpt350to500_mcstat_bin2
141 CMS_HWW_boosted_jmr_sig_2016APV
142 CMS_HWW_boosted_jmr_ttbar_2016APV
143 CMS_scale_j_Abs
144 CMS_HWW_boosted_jmr_ttbar_2016
145 CMS_HWW_boosted_jms_ttbar_2016APV
146 QCDscale_single_top
147 CMS_scale_j_BBEC1_2018
148 CMS_HWW_boosted_VBF_mcstat_bin4
149 lumi_13TeV_correlated
150 CMS_scale_j_BBEC1_2016APV
−2
−1
0
1
2
−0.05
0
0.05
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$
$(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$

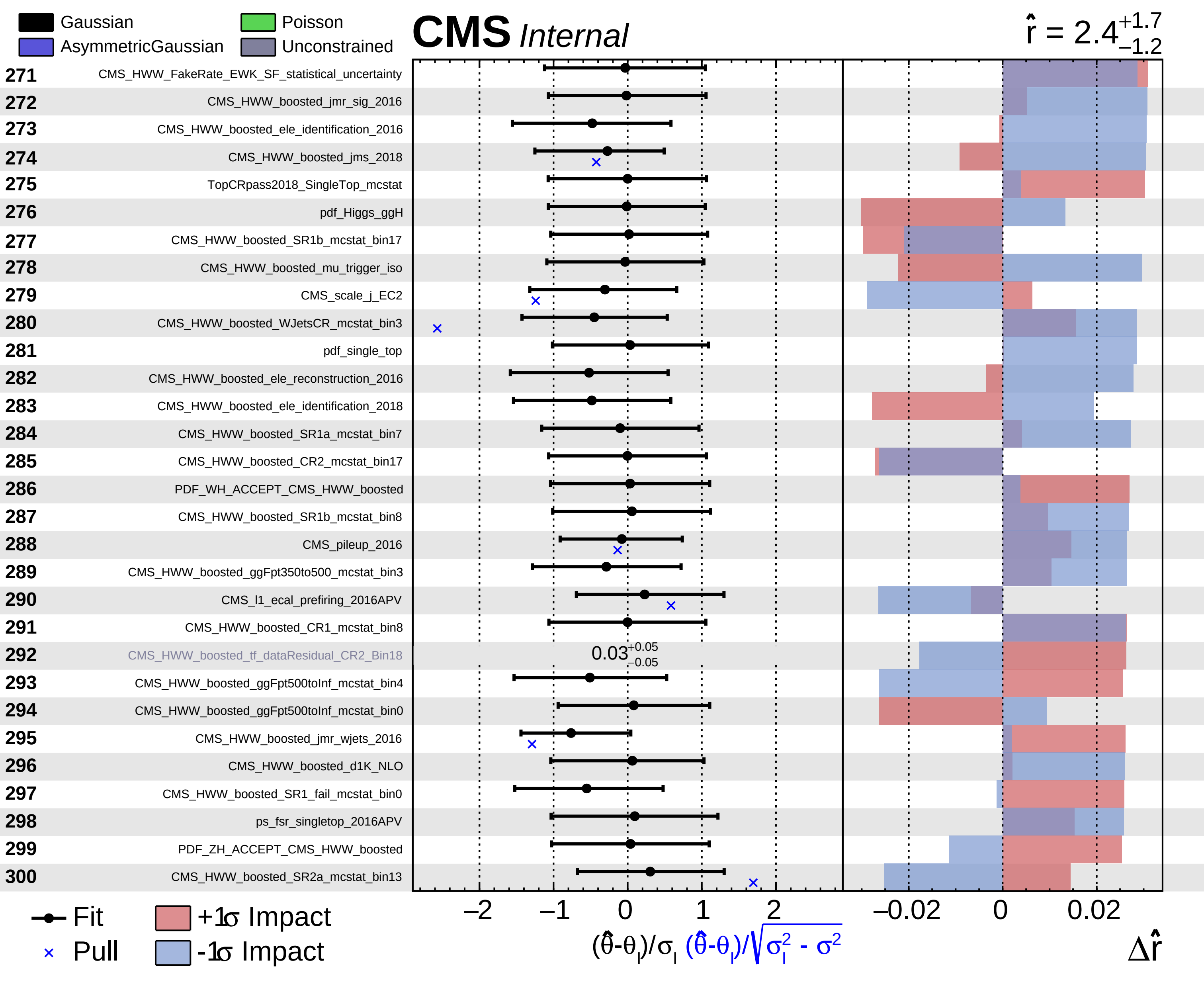


Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Internal
$\hat{r} = 2.4^{+1.7}_{-1.2}$
151 CMS_HWW_boosted_btagSFbc_2018
152 ps_fsr_ttH
153 CMS_HWW_boosted_TopCR_mcstat_bin5
154 CMS_HWW_boosted_jms_2016APV
155 CMS_HWW_boosted_SR2a_mcstat_bin3
156 CMS_res_j_2016
157 CMS_scale_j_BBEC1_2016
158 ps_fsr_singletop
159 CMS_lumi_13TeV_2018
160 CMS_HWW_boosted_TopCR_mcstat_bin7
161 CMS_HWW_boosted_tf_dataResidual_1b_MH_Reco_par0
$0.970^{+0}_{-0.005}$
162 CMS_HWW_boosted_ggFpt250to350_mcstat_bin1
163 TopCRpass2017_TTbar_mcstat
164 CMS_HWW_boosted_electron_veto
165 CMS_HWW_boosted_ggFpt250to350_mcstat_bin3
166 CMS_res_j_2016APV
167 CMS_HWW_boosted_mu_identification
168 CMS_HWW_boosted_TopCR_mcstat_bin6
169 CMS_scale_j_EC2_2016
170 CMS_HWW_boosted_SR1_pass_mcstat_bin3
171 CMS_HWW_boosted_ggFpt350to500_mcstat_bin0
172 CMS_HWW_boosted_SR2a_mcstat_bin4
173 CMS_scale_j_HF_2016APV
174 ps_fsr_ttbar
175 CMS_scale_j_FlavQCD
176 PDF_single_top_ACCEPT_CMS_HWW_boosted
177 CMS_HWW_boosted_tf_dataResidual_2b_MH_Reco_par1
$0.96^{+0}_{-0.13}$
178 pdf_ttbar
179 CMS_scale_j_BBEC1
180 CMS_HWW_boosted_FakeRate_EWK_SF_statistical_uncertainty
−2
−1
0
1
2
−0.05
0
0.05
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$ $(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$

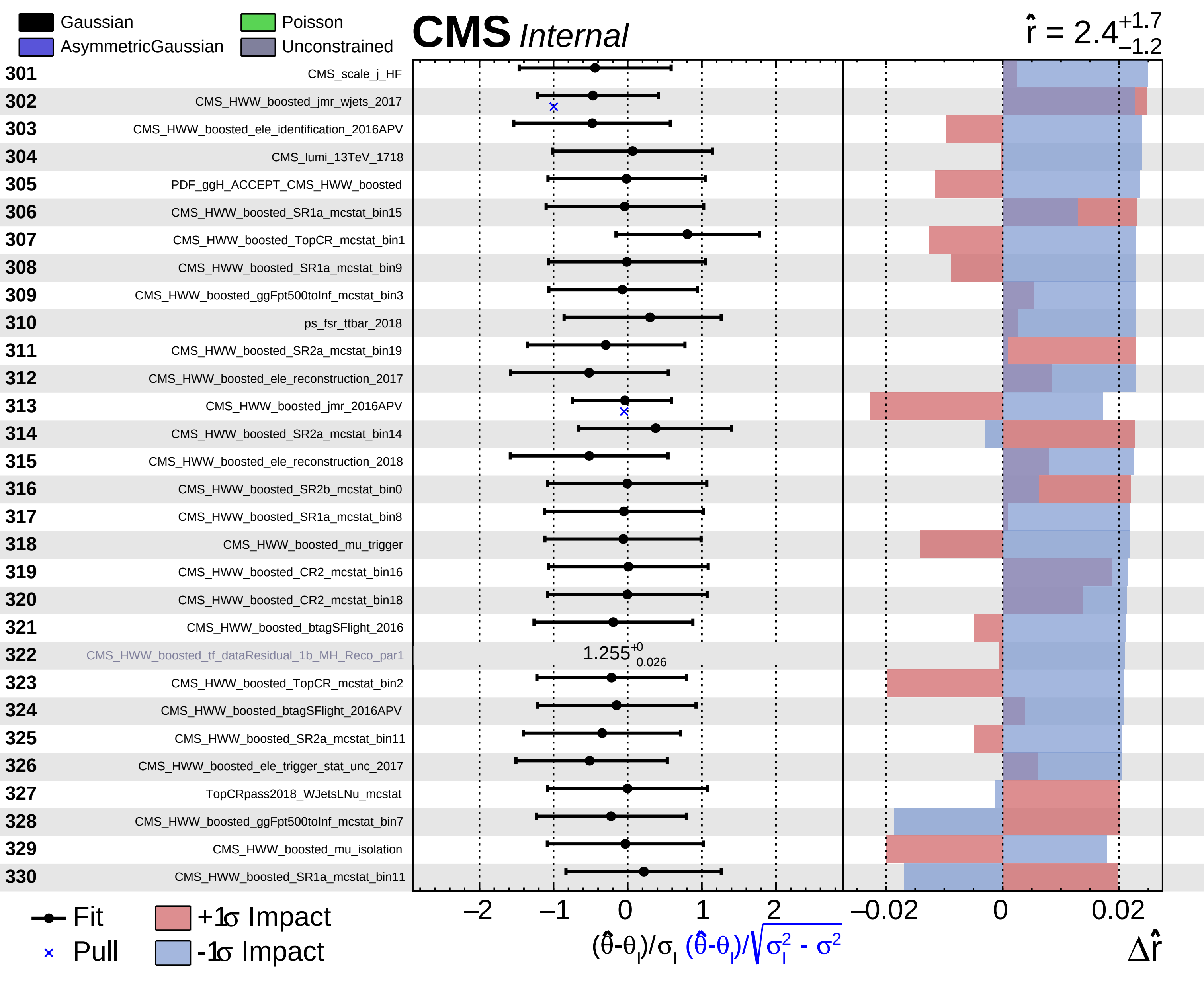

Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Internal
r̂ = 2.4 +1.7 −1.2
181 CMS_HWW_boosted_triggerEffSF_uncorrelated
182 CMS_scale_j_Abs_2016APV
183 CMS_scale_j_HF_2016
184 BR_hww
185 CMS_HWW_boosted_btagSFbcCorrelated
186 QCDscale_singletop_ACCEPT_CMS_HWW_boosted
187 CMS_HWW_boosted_SR1_pass_mcstat_bin2
188 CMS_HWW_boosted_btagSFbc_2016APV
189 CMS_HWW_boosted_tf_dataResidual_2b_MH_Reco_par2
1.37 +0.10 −0.27
190 CMS_HWW_boosted_SR2a_mcstat_bin10
191 CMS_HWW_boosted_ggFpt250to350_mcstat_bin7
192 CMS_HWW_boosted_ggFpt250to350_mcstat_bin2
193 CMS_HWW_boosted_btagSFlight_2017
194 pdf_rest_bkg
195 CMS_HWW_boosted_ele_identification
196 CMS_HWW_boosted_tf_dataResidual_2b_MH_Reco_par3
0.3 +0.8 −1.3
197 CMS_HWW_boosted_TopCR_mcstat_bin4
198 CMS_res_j_2017
199 PDF_wjets_ACCEPT_CMS_HWW_boosted
200 QCDscale_WH_ACCEPT_CMS_HWW_boosted
201 ps_isr_qqH
202 TopCRpass2018_TTbar_mcstat
203 ps_fsr_singletop_2018
204 CMS_HWW_boosted_ele_trigger_stat_unc_2016
205 CMS_HWW_boosted_tf_dataResidual_2b_MH_Reco_par0
1.10 +0 −0.05
206 CMS_HWW_boosted_ggFpt350to500_mcstat_bin4
207 CMS_HWW_boosted_VBF_mcstat_bin5
208 CMS_HWW_boosted_ggFpt350to500_mcstat_bin7
209 CMS_HWW_boosted_SR1a_mcstat_bin3
210 CMS_HWW_boosted_tf_dataResidual_1b_MH_Reco_par6
1.806 +0 −0.019
−2
−1
0
1
2
−0.05
0
0.05
Fit
Pull
+1σ Impact
−1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$ $(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$


Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Internal
r̂ = 2.4 +1.7 −1.2
211 CMS_HWW_boosted_rest_bkg_rate
212 ps_fsr_ttbar_2016APV
213 ps_fsr_single_top
214 CMS_HWW_boosted_FakeRate_statistical_uncertainty
215 QCDscale_ttbar
216 CMS_HWW_boosted_SR2b_mcstat_bin16
217 ps_fsr_ttbar_2016
218 ps_isr_ggH
219 CMS_HWW_boosted_btagSFbc_2017
220 ps_fsr_ttbar_2017
221 QCDscale_ttbar_ACCEPT_CMS_HWW_boosted
222 CMS_HWW_boosted_SR2a_single_top_mcstat_bin18
223 CMS_HWW_boosted_wjets_rate
224 CMS_lumi_13TeV_2017
225 lumi_13TeV_2016
226 CMS_HWW_boosted_SR1a_mcstat_bin13
227 CMS_HWW_boosted_muon_veto
228 CMS_HWW_boosted_ggFpt250to350_mcstat_bin5
229 alpha_s
230 CMS_HWW_boosted_ggFpt350to500_mcstat_bin5
231 CMS_HWW_boosted_tf_dataResidual_2b_MH_Reco_par5
1.34 +0.23 −0.31
232 CMS_HWW_boosted_d2K_NLO
233 CMS_HWW_boosted_SR1a_mcstat_bin4
234 CMS_HWW_boosted_W_d1kappa_EW
235 CMS_HWW_boosted_jms_sig_2016
236 CMS_HWW_boosted_tf_dataResidual_1b_MH_Reco_par2
0.85 +0 −0.07
237 ps_fsr_wjets_2016
238 CMS_HWW_boosted_ele_reconstruction
239 CMS_HWW_boosted_ggFpt350to500_mcstat_bin6
240 CMS_HWW_boosted_ggFpt500toInf_mcstat_bin1
−2 −1 0 1 2
−0.04 −0.02 0 0.02 0.04
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$ $(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$

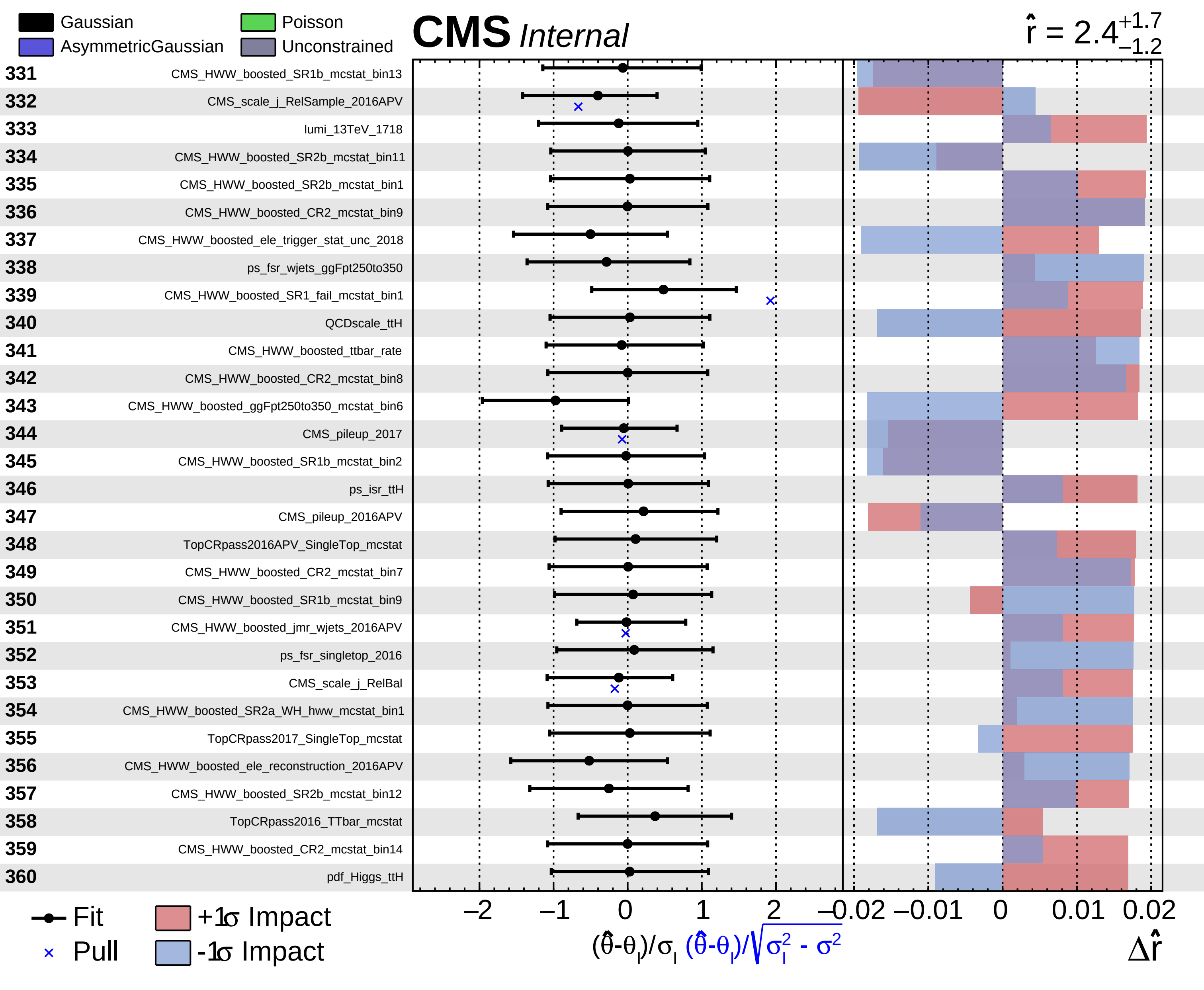

Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Internal
$\hat{r} = 2.4^{+1.7}_{-1.2}$
241 ps_isr_rest_bkg
242 CMS_HWW_boosted_SR2a_mcstat_bin17
243 CMS_HWW_boosted_ggFpt500toInf_mcstat_bin6
244 CMS_HWW_boosted_SR1_fail_mcstat_bin3
245 CMS_HWW_boosted_ele_trigger_syst_unc
246 wjetsnormSF
$1.12^{+0.20}_{-0.13}$
247 ps_isr_single_top
248 CMS_HWW_boosted_SR2b_mcstat_bin7
249 CMS_HWW_boosted_TopCR_mcstat_bin3
250 CMS_l1_ecal_prefiring_2016
251 CMS_HWW_boosted_ggFpt500toInf_mcstat_bin5
252 ps_fsr_wjets_2017
253 ps_fsr_wjets_ggFpt350to500
254 CMS_HWW_boosted_ggFpt250to350_mcstat_bin0
255 ps_fsr_wjets_ggFpt500toInf
256 CMS_HWW_boosted_btagSFbc_2016
257 CMS_HWW_boosted_SR2a_mcstat_bin12
258 CMS_HWW_boosted_SR1_fail_mcstat_bin2
259 CMS_HWW_boosted_tf_dataResidual_2b_MH_Reco_par6
$1.88^{+0.13}_{-0.11}$
260 CMS_HWW_boosted_SR2b_mcstat_bin17
261 CMS_HWW_boosted_btagSFlight_2018
262 CMS_HWW_boosted_TopCR_mcstat_bin0
263 CMS_HWW_boosted_W_d3kappa_EW
264 lumi_13TeV_2018
265 CMS_HWW_boosted_SR1a_mcstat_bin1
266 CMS_HWW_boosted_SR1a_mcstat_bin10
267 CMS_HWW_boosted_SR2a_mcstat_bin15
268 CMS_HWW_boosted_SR2b_mcstat_bin15
269 CMS_res_j_2018
270 CMS_HWW_boosted_SR2a_mcstat_bin2
−2
−1
0
1
2
−0.04
−0.02
0
0.02
0.04
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$
$(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$

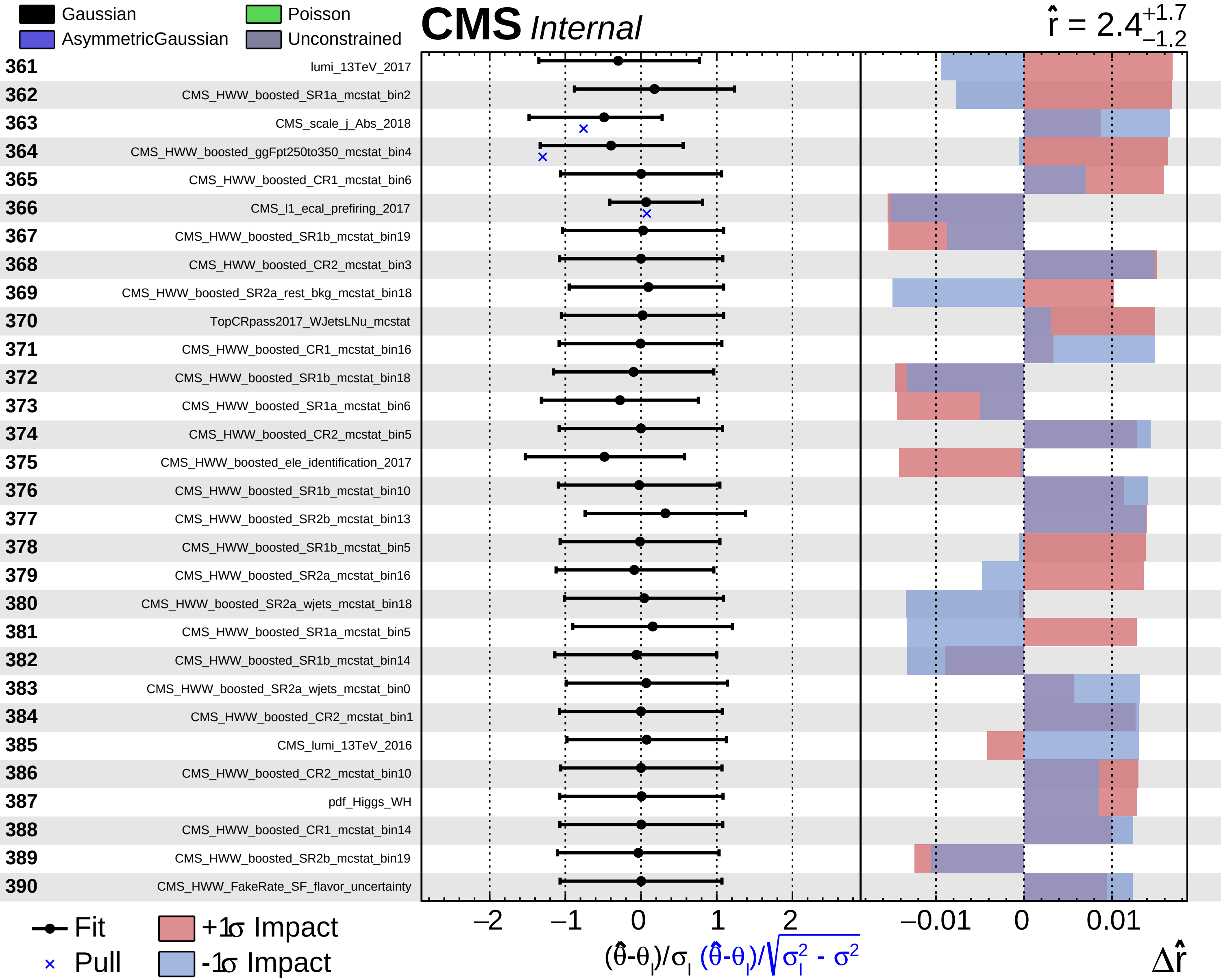


Gaussian
AsymmetricGaussian
Poisson
Unconstrained
CMS Internal
r̂ = 2.4 +1.7 −1.2
271 CMS_HWW_FakeRate_EWK_SF_statistical_uncertainty
272 CMS_HWW_boosted_jmr_sig_2016
273 CMS_HWW_boosted_ele_identification_2016
274 CMS_HWW_boosted_jms_2018
275 TopCRpass2018_SingleTop_mcstat
276 pdf_Higgs_ggH
277 CMS_HWW_boosted_SR1b_mcstat_bin17
278 CMS_HWW_boosted_mu_trigger_iso
279 CMS_scale_j_EC2
280 CMS_HWW_boosted_WJetsCR_mcstat_bin3
281 pdf_single_top
282 CMS_HWW_boosted_ele_reconstruction_2016
283 CMS_HWW_boosted_ele_identification_2018
284 CMS_HWW_boosted_SR1a_mcstat_bin7
285 CMS_HWW_boosted_CR2_mcstat_bin17
286 PDF_WH_ACCEPT_CMS_HWW_boosted
287 CMS_HWW_boosted_SR1b_mcstat_bin8
288 CMS_pileup_2016
289 CMS_HWW_boosted_ggFpt350to500_mcstat_bin3
290 CMS_l1_ecal_prefiring_2016APV
291 CMS_HWW_boosted_CR1_mcstat_bin8
292 CMS_HWW_boosted_tf_dataResidual_CR2_Bin18
0.03 +0.05 −0.05
293 CMS_HWW_boosted_ggFpt500toInf_mcstat_bin4
294 CMS_HWW_boosted_ggFpt500toInf_mcstat_bin0
295 CMS_HWW_boosted_jmr_wjets_2016
296 CMS_HWW_boosted_d1K_NLO
297 CMS_HWW_boosted_SR1_fail_mcstat_bin0
298 ps_fsr_singletop_2016APV
299 PDF_ZH_ACCEPT_CMS_HWW_boosted
300 CMS_HWW_boosted_SR2a_mcstat_bin13
−2
−1
0
1
2
−0.02
0
0.02
Fit
Pull
+1σ Impact
-1σ Impact
(θ̂-θ_I)/σ_I (θ̂-θ_I)/√(σ_I² - σ²)
Δr̂

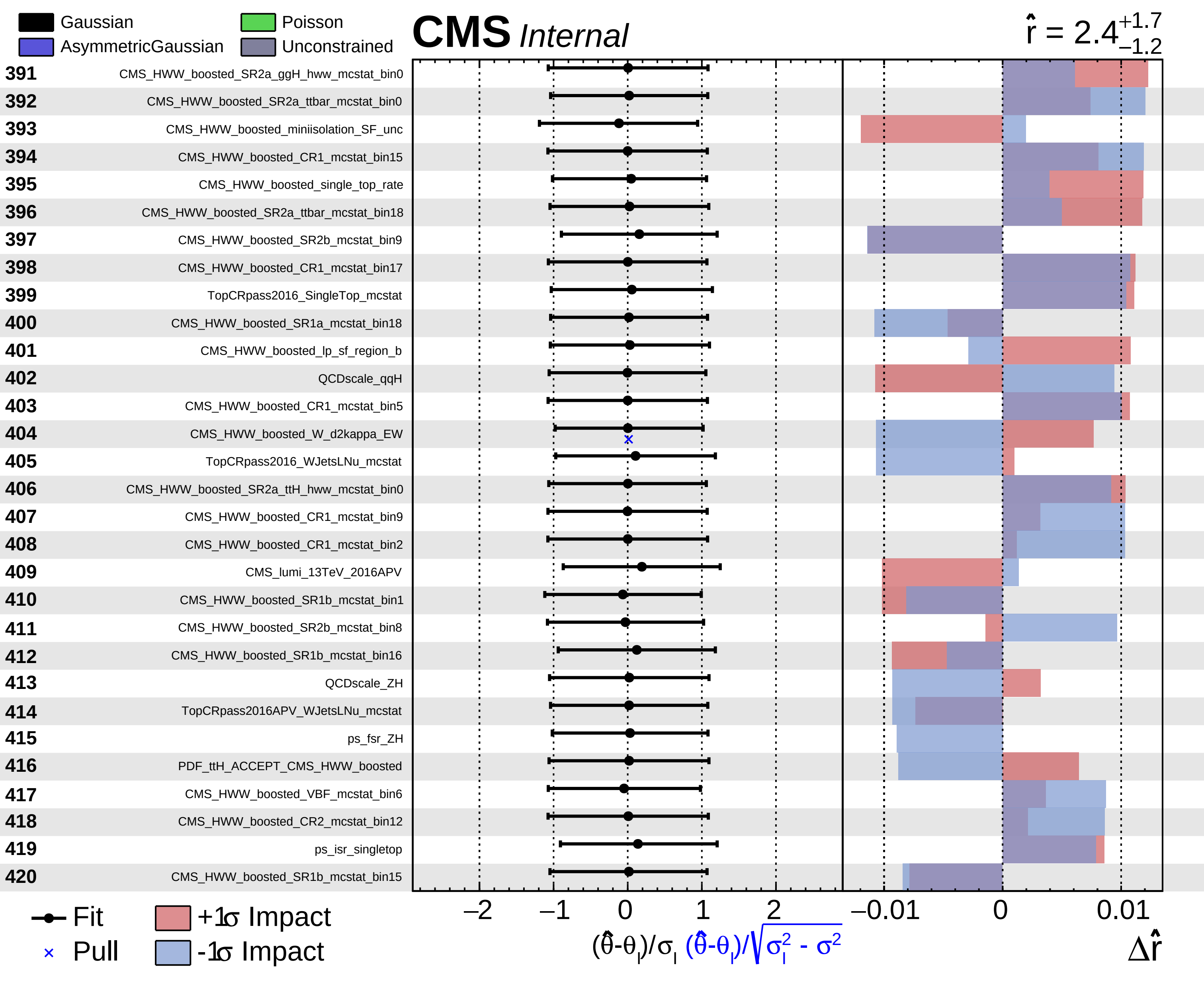


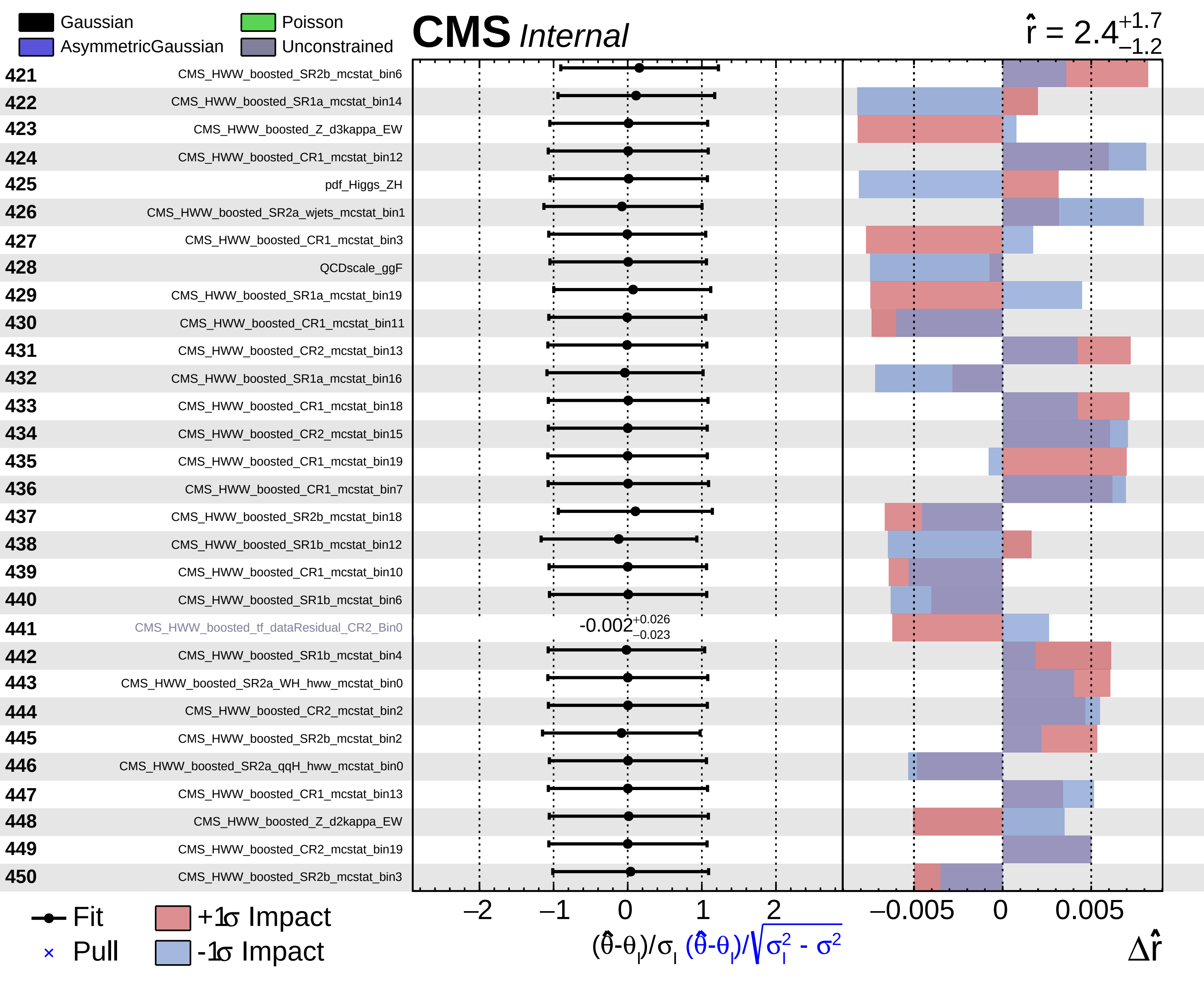
Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Internal
$\hat{r} = 2.4^{+1.7}_{-1.2}$
301 CMS_scale_j_HF
302 CMS_HWW_boosted_jmr_wjets_2017
303 CMS_HWW_boosted_ele_identification_2016APV
304 CMS_lumi_13TeV_1718
305 PDF_ggH_ACCEPT_CMS_HWW_boosted
306 CMS_HWW_boosted_SR1a_mcstat_bin15
307 CMS_HWW_boosted_TopCR_mcstat_bin1
308 CMS_HWW_boosted_SR1a_mcstat_bin9
309 CMS_HWW_boosted_ggFpt500toInf_mcstat_bin3
310 ps_fsr_ttbar_2018
311 CMS_HWW_boosted_SR2a_mcstat_bin19
312 CMS_HWW_boosted_ele_reconstruction_2017
313 CMS_HWW_boosted_jmr_2016APV
314 CMS_HWW_boosted_SR2a_mcstat_bin14
315 CMS_HWW_boosted_ele_reconstruction_2018
316 CMS_HWW_boosted_SR2b_mcstat_bin0
317 CMS_HWW_boosted_SR1a_mcstat_bin8
318 CMS_HWW_boosted_mu_trigger
319 CMS_HWW_boosted_CR2_mcstat_bin16
320 CMS_HWW_boosted_CR2_mcstat_bin18
321 CMS_HWW_boosted_btagSFlight_2016
322 CMS_HWW_boosted_tf_dataResidual_1b_MH_Reco_par1
$1.255^{+0}_{-0.026}$
323 CMS_HWW_boosted_TopCR_mcstat_bin2
324 CMS_HWW_boosted_btagSFlight_2016APV
325 CMS_HWW_boosted_SR2a_mcstat_bin11
326 CMS_HWW_boosted_ele_trigger_stat_unc_2017
327 TopCRpass2018_WJetsLNu_mcstat
328 CMS_HWW_boosted_ggFpt500toInf_mcstat_bin7
329 CMS_HWW_boosted_mu_isolation
330 CMS_HWW_boosted_SR1a_mcstat_bin11
−2
−1
0
1
2
−0.02
0
0.02
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$ $(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$


Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Internal
$\hat{r} = 2.4^{+1.7}_{-1.2}$
331 CMS_HWW_boosted_SR1b_mcstat_bin13
332 CMS_scale_j_RelSample_2016APV
333 lumi_13TeV_1718
334 CMS_HWW_boosted_SR2b_mcstat_bin11
335 CMS_HWW_boosted_SR2b_mcstat_bin1
336 CMS_HWW_boosted_CR2_mcstat_bin9
337 CMS_HWW_boosted_ele_trigger_stat_unc_2018
338 ps_fsr_wjets_ggFpt250to350
339 CMS_HWW_boosted_SR1_fail_mcstat_bin1
340 QCDscale_ttH
341 CMS_HWW_boosted_ttbar_rate
342 CMS_HWW_boosted_CR2_mcstat_bin8
343 CMS_HWW_boosted_ggFpt250to350_mcstat_bin6
344 CMS_pileup_2017
345 CMS_HWW_boosted_SR1b_mcstat_bin2
346 ps_isr_ttH
347 CMS_pileup_2016APV
348 TopCRpass2016APV_SingleTop_mcstat
349 CMS_HWW_boosted_CR2_mcstat_bin7
350 CMS_HWW_boosted_SR1b_mcstat_bin9
351 CMS_HWW_boosted_jmr_wjets_2016APV
352 ps_fsr_singletop_2016
353 CMS_scale_j_RelBal
354 CMS_HWW_boosted_SR2a_WH_hww_mcstat_bin1
355 TopCRpass2017_SingleTop_mcstat
356 CMS_HWW_boosted_ele_reconstruction_2016APV
357 CMS_HWW_boosted_SR2b_mcstat_bin12
358 TopCRpass2016_TTbar_mcstat
359 CMS_HWW_boosted_CR2_mcstat_bin14
360 pdf_Higgs_ttH
−2
−1
0
1
2
−0.02
−0.01
0
0.01
0.02
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$ $(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$

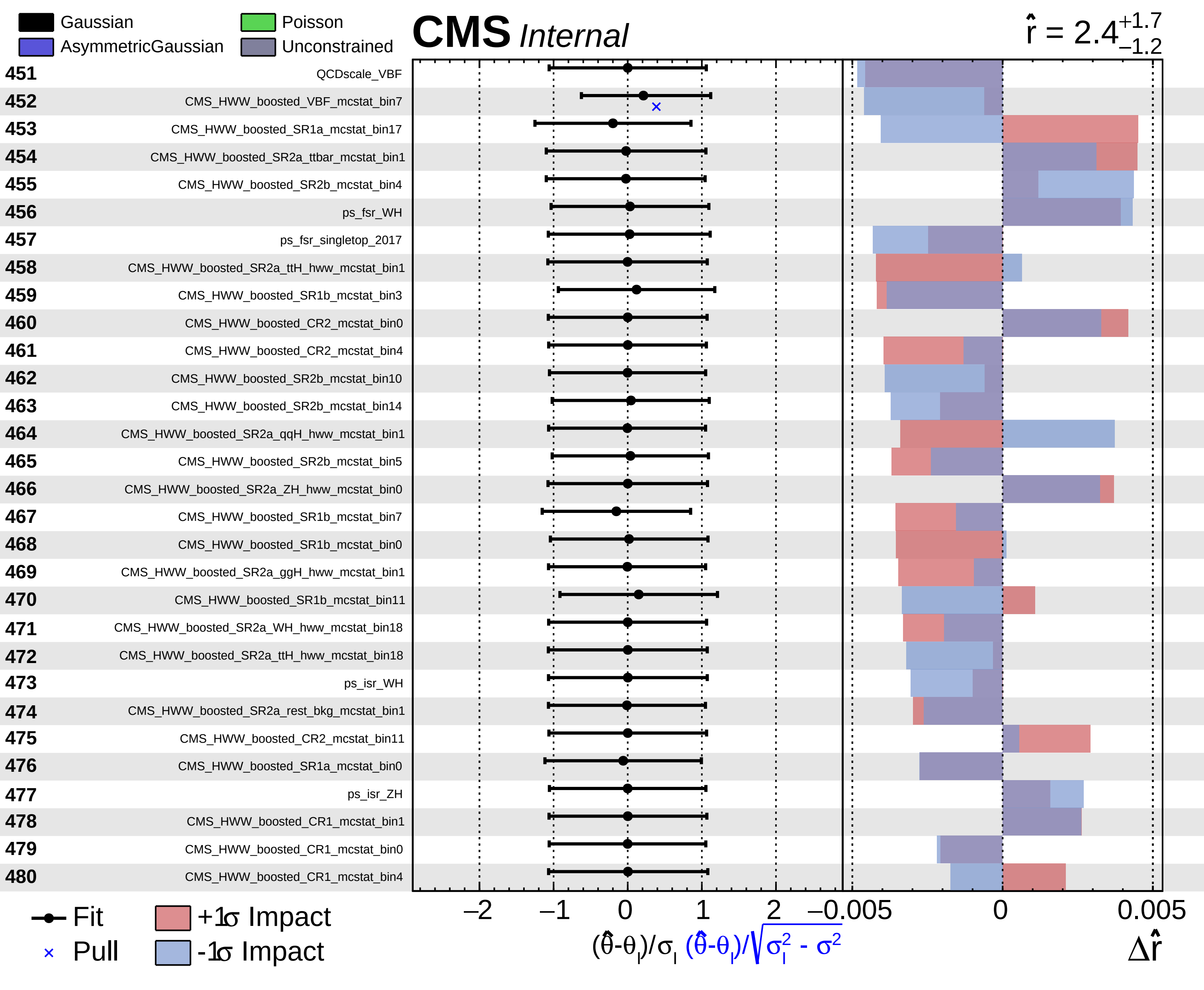


Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Internal
$\hat{r} = 2.4^{+1.7}_{-1.2}$
361 lumi_13TeV_2017
362 CMS_HWW_boosted_SR1a_mcstat_bin2
363 CMS_scale_j_Abs_2018
364 CMS_HWW_boosted_ggFpt250to350_mcstat_bin4
365 CMS_HWW_boosted_CR1_mcstat_bin6
366 CMS_l1_ecal_prefiring_2017
367 CMS_HWW_boosted_SR1b_mcstat_bin19
368 CMS_HWW_boosted_CR2_mcstat_bin3
369 CMS_HWW_boosted_SR2a_rest_bkg_mcstat_bin18
370 TopCRpass2017_WJetsLNu_mcstat
371 CMS_HWW_boosted_CR1_mcstat_bin16
372 CMS_HWW_boosted_SR1b_mcstat_bin18
373 CMS_HWW_boosted_SR1a_mcstat_bin6
374 CMS_HWW_boosted_CR2_mcstat_bin5
375 CMS_HWW_boosted_ele_identification_2017
376 CMS_HWW_boosted_SR1b_mcstat_bin10
377 CMS_HWW_boosted_SR2b_mcstat_bin13
378 CMS_HWW_boosted_SR1b_mcstat_bin5
379 CMS_HWW_boosted_SR2a_mcstat_bin16
380 CMS_HWW_boosted_SR2a_wjets_mcstat_bin18
381 CMS_HWW_boosted_SR1a_mcstat_bin5
382 CMS_HWW_boosted_SR1b_mcstat_bin14
383 CMS_HWW_boosted_SR2a_wjets_mcstat_bin0
384 CMS_HWW_boosted_CR2_mcstat_bin1
385 CMS_lumi_13TeV_2016
386 CMS_HWW_boosted_CR2_mcstat_bin10
387 pdf_Higgs_WH
388 CMS_HWW_boosted_CR1_mcstat_bin14
389 CMS_HWW_boosted_SR2b_mcstat_bin19
390 CMS_HWW_FakeRate_SF_flavor_uncertainty
−2
−1
0
1
2
−0.01
0
0.01
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$ $(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$

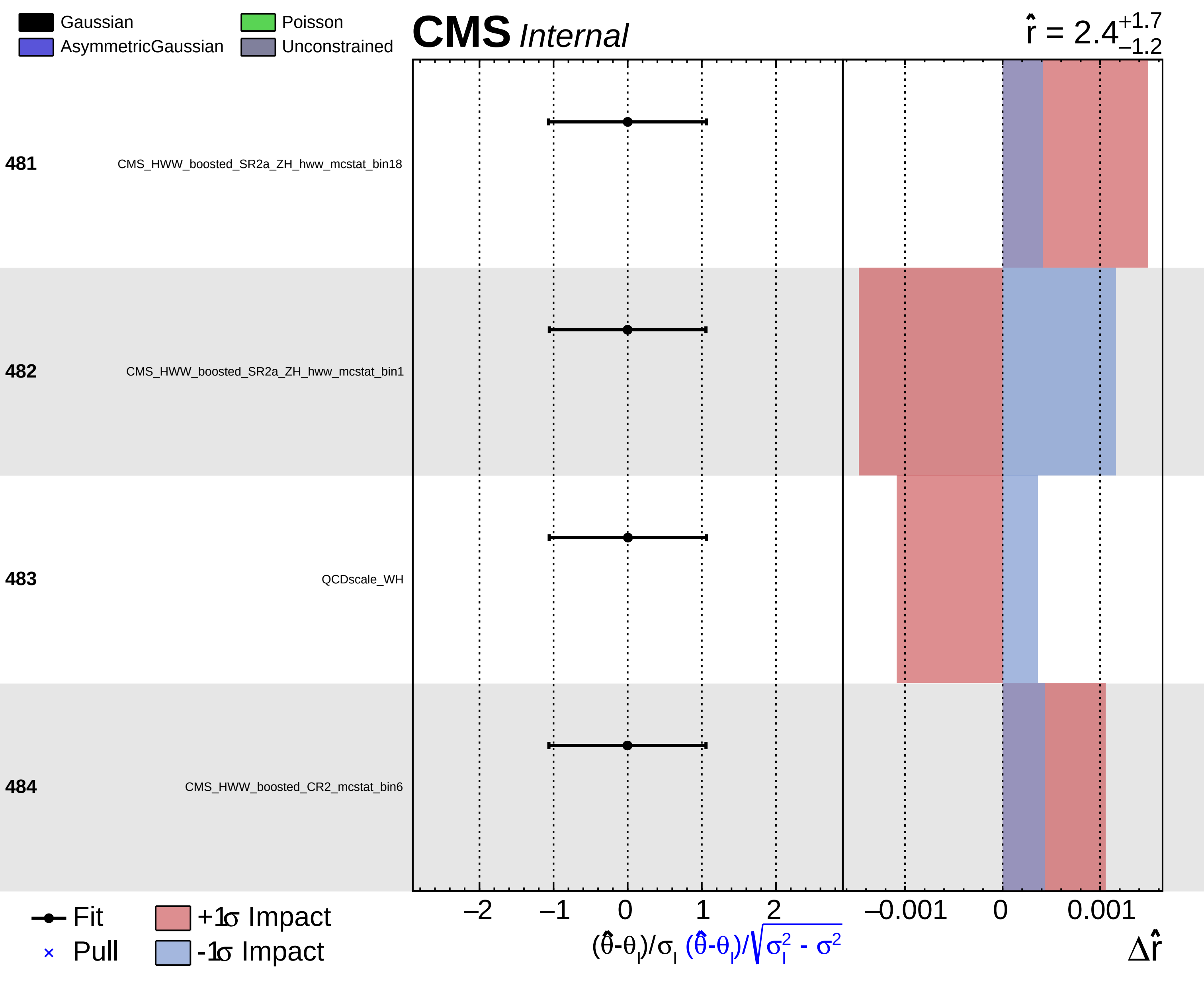

Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Internal
$\hat{r} = 2.4^{+1.7}_{-1.2}$
391 CMS_HWW_boosted_SR2a_ggH_hww_mcstat_bin0
392 CMS_HWW_boosted_SR2a_ttbar_mcstat_bin0
393 CMS_HWW_boosted_miniisolation_SF_unc
394 CMS_HWW_boosted_CR1_mcstat_bin15
395 CMS_HWW_boosted_single_top_rate
396 CMS_HWW_boosted_SR2a_ttbar_mcstat_bin18
397 CMS_HWW_boosted_SR2b_mcstat_bin9
398 CMS_HWW_boosted_CR1_mcstat_bin17
399 TopCRpass2016_SingleTop_mcstat
400 CMS_HWW_boosted_SR1a_mcstat_bin18
401 CMS_HWW_boosted_lp_sf_region_b
402 QCDscale_qqH
403 CMS_HWW_boosted_CR1_mcstat_bin5
404 CMS_HWW_boosted_W_d2kappa_EW
405 TopCRpass2016_WJetsLNu_mcstat
406 CMS_HWW_boosted_SR2a_ttH_hww_mcstat_bin0
407 CMS_HWW_boosted_CR1_mcstat_bin9
408 CMS_HWW_boosted_CR1_mcstat_bin2
409 CMS_lumi_13TeV_2016APV
410 CMS_HWW_boosted_SR1b_mcstat_bin1
411 CMS_HWW_boosted_SR2b_mcstat_bin8
412 CMS_HWW_boosted_SR1b_mcstat_bin16
413 QCDscale_ZH
414 TopCRpass2016APV_WJetsLNu_mcstat
415 ps_fsr_ZH
416 PDF_ttH_ACCEPT_CMS_HWW_boosted
417 CMS_HWW_boosted_VBF_mcstat_bin6
418 CMS_HWW_boosted_CR2_mcstat_bin12
419 ps_isr_singletop
420 CMS_HWW_boosted_SR1b_mcstat_bin15
−2 −1 0 1 2
−0.01 0 0.01
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$ $(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$



Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Internal
$\hat{r} = 2.4^{+1.7}_{-1.2}$
421 CMS_HWW_boosted_SR2b_mcstat_bin6
422 CMS_HWW_boosted_SR1a_mcstat_bin14
423 CMS_HWW_boosted_Z_d3kappa_EW
424 CMS_HWW_boosted_CR1_mcstat_bin12
425 pdf_Higgs_ZH
426 CMS_HWW_boosted_SR2a_wjets_mcstat_bin1
427 CMS_HWW_boosted_CR1_mcstat_bin3
428 QCDscale_ggF
429 CMS_HWW_boosted_SR1a_mcstat_bin19
430 CMS_HWW_boosted_CR1_mcstat_bin11
431 CMS_HWW_boosted_CR2_mcstat_bin13
432 CMS_HWW_boosted_SR1a_mcstat_bin16
433 CMS_HWW_boosted_CR1_mcstat_bin18
434 CMS_HWW_boosted_CR2_mcstat_bin15
435 CMS_HWW_boosted_CR1_mcstat_bin19
436 CMS_HWW_boosted_CR1_mcstat_bin7
437 CMS_HWW_boosted_SR2b_mcstat_bin18
438 CMS_HWW_boosted_SR1b_mcstat_bin12
439 CMS_HWW_boosted_CR1_mcstat_bin10
440 CMS_HWW_boosted_SR1b_mcstat_bin6
441 CMS_HWW_boosted_tf_dataResidual_CR2_Bin0
$-0.002^{+0.026}_{-0.023}$
442 CMS_HWW_boosted_SR1b_mcstat_bin4
443 CMS_HWW_boosted_SR2a_WH_hww_mcstat_bin0
444 CMS_HWW_boosted_CR2_mcstat_bin2
445 CMS_HWW_boosted_SR2b_mcstat_bin2
446 CMS_HWW_boosted_SR2a_qqH_hww_mcstat_bin0
447 CMS_HWW_boosted_CR1_mcstat_bin13
448 CMS_HWW_boosted_Z_d2kappa_EW
449 CMS_HWW_boosted_CR2_mcstat_bin19
450 CMS_HWW_boosted_SR2b_mcstat_bin3
−2 −1 0 1 2
−0.005 0 0.005
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$ $(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$



Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Internal
$\hat{r} = 2.4^{+1.7}_{-1.2}$
451 QCDscale_VBF
452 CMS_HWW_boosted_VBF_mcstat_bin7
453 CMS_HWW_boosted_SR1a_mcstat_bin17
454 CMS_HWW_boosted_SR2a_ttbar_mcstat_bin1
455 CMS_HWW_boosted_SR2b_mcstat_bin4
456 ps_fsr_WH
457 ps_fsr_singletop_2017
458 CMS_HWW_boosted_SR2a_ttH_hww_mcstat_bin1
459 CMS_HWW_boosted_SR1b_mcstat_bin3
460 CMS_HWW_boosted_CR2_mcstat_bin0
461 CMS_HWW_boosted_CR2_mcstat_bin4
462 CMS_HWW_boosted_SR2b_mcstat_bin10
463 CMS_HWW_boosted_SR2b_mcstat_bin14
464 CMS_HWW_boosted_SR2a_qqH_hww_mcstat_bin1
465 CMS_HWW_boosted_SR2b_mcstat_bin5
466 CMS_HWW_boosted_SR2a_ZH_hww_mcstat_bin0
467 CMS_HWW_boosted_SR1b_mcstat_bin7
468 CMS_HWW_boosted_SR1b_mcstat_bin0
469 CMS_HWW_boosted_SR2a_ggH_hww_mcstat_bin1
470 CMS_HWW_boosted_SR1b_mcstat_bin11
471 CMS_HWW_boosted_SR2a_WH_hww_mcstat_bin18
472 CMS_HWW_boosted_SR2a_ttH_hww_mcstat_bin18
473 ps_isr_WH
474 CMS_HWW_boosted_SR2a_rest_bkg_mcstat_bin1
475 CMS_HWW_boosted_CR2_mcstat_bin11
476 CMS_HWW_boosted_SR1a_mcstat_bin0
477 ps_isr_ZH
478 CMS_HWW_boosted_CR1_mcstat_bin1
479 CMS_HWW_boosted_CR1_mcstat_bin0
480 CMS_HWW_boosted_CR1_mcstat_bin4
Fit
Pull
+1σ Impact
-1σ Impact
−2
−1
0
1
2
−0.005
0
0.005
$(\hat{\theta}-\theta_I)/\sigma_I$
$(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$


Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Internal
$\hat{r} = 2.4^{+1.7}_{-1.2}$
481 CMS_HWW_boosted_SR2a_ZH_hww_mcstat_bin18
482 CMS_HWW_boosted_SR2a_ZH_hww_mcstat_bin1
483 QCDscale_WH
484 CMS_HWW_boosted_CR2_mcstat_bin6
−2 −1 0 1 2
−0.001 0 0.001
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$ $(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$